

# OSBORNE V. LOPES

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## SUMMARY

Driven Data Graduate with experience building and optimizing ETL pipelines using Python, SQL, AWS, Airflow and Snowflake. Expertise in data warehousing, batch & streaming processing, and scalable analytics platforms for business insights.

## EDUCATION

### Northeastern University, USA

Dec 2025

*Master of Science, Information Systems*

- **Coursework:** Big Data and Intelligent Analytics, Data Architecture & Business Intelligence, Data Science Methods & Tools

### University of Mumbai, India

Jun 2021

*Bachelor of Technology, Information Technology*

- **Coursework:** Data Mining and Business Intelligence, Database Management, Cloud Computing Services

## SKILLS

- **Databases & Cloud:** Snowflake, SQL, Git, MongoDB, DynamoDB, S3, Glue, Redshift, EMR, Lambda, Athena, IAM, Kinesis
- **Languages:** Python, R, SQL, Pro\*C
- **Tools:** Airflow, ADF, Control-M, DBT, Apache Spark, Kafka, Databricks
- **Cloud Platforms:** AWS (S3, Glue, EMR, Lambda, Athena, Kinesis, IAM), Azure (ADF, Blob Storage), Docker
- **Analytical Toolkit:** Tableau, PowerBI, Google Data Studio, Excel, QuickSight
- **AI Data Automation:** RAG, Agentic AI, PineCone, LLM, n8n, Prompt engineering

## EXPERIENCE

### Atlas SP | *Data Engineer Co-op* | *New York, NY*

Jan 2025 - Aug 2025

- Implemented ADF pipelines to automate ingestion of millions of records from multiple sources, reducing processing time
- Stored processed data in Azure Blob Storage & vector indexes, enabling RAG-based querying improving consistency by 30%.
- Designed normalized & dimensional data models in Snowflake supporting analytics, improving query performance and reporting.
- Refactored and optimized queries in Snowflake to retrieve data only on market days, resulting in cleaner and faster data retrieval
- Optimized batch streaming workflows using Databricks, integrating Kafka & Presto for scalable data processing, reducing latency

### Ingram Micro | *Business Intelligence Analyst* | *Mumbai, India*

Oct 2022 - Aug 2023

- Designed Power BI semantic models using star-schema data modeling and advanced DAX measures to track logistics KPIs
- Modularized transformations and incremental DBT models, reducing data processing time by ~25% and accelerating OPS analytics.
- Engineered Fault-tolerant ETL pipelines to automate tasks in IMFIRST and improved workflows, boosting efficiency by 30%
- Developed QuickSight dashboards, enabling operations teams to monitor shipment status, transit delays, and SLA adherence
- Improved data freshness and reporting reliability by automating ingestion scheduling and retries via Airflow within CI/CD pipelines
- Built reusable SQL models to support analytics and reporting, owning end-to-end dataset reliability for 5+ business teams

### ICICI Securities | *Associate Software Developer* | *Mumbai, India*

Jun 2021 - Sep 2022

- Led 5 Disaster Recovery (DR) drills, ensuring uninterrupted connectivity to NSE & BSE trading platforms, mitigating potential loss
- Optimized complex SQL queries using window functions and partition pruning, reducing time by 40%, enabling faster reporting
- Deployed Python scripts for data cleaning, reducing processing time by 50% & automated Tableau reports, saving 8 hours weekly
- Addressed and resolved 20+ live issues during market hours, maintaining uninterrupted operations of critical systems

## PROJECTS

### NewsScope | [GitHub](#)

Apr 2024

- Built NewsScope, an industry-specific news aggregator, using Python (BeautifulSoup & Scrapy), Snowflake & DBT
- Maintained a robust data schema, performing rigorous data validations (Pydantic 2 & Pytest) to guarantee high-quality data
- Designed a user-friendly Streamlit interface & automated ETL pipelines with Airflow, containerizing the platform with Docker

### Drug Review Analysis for Patient Condition Diagnosis

Sep 2023

- Crafted an NLP tool for classifying patient conditions from drug reviews, implementing data lemmatization & stemming
- Visualized sentiment distribution with pie charts to enhance data interpretation & tested various models achieving high accuracy

### HR Analytics

Jul 2023

- Created an HR analytics dashboard for visualizing and analyzing employee performance and HR metrics
- Cleaned and prepared data in Microsoft Excel & designed interactive dashboards with advanced DAX calculations